A

PROJECT BASED LEARNING

ON

RCC LIFT PIT PLAN DESIGN

BY

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UNDER THE GUIDANCE OF

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ABSTRACT

An RCC Lift Pit is a reinforced cement concrete structure constructed below the lowest lift landing level to accommodate the required lift equipment and provide the necessary structural enclosure. The lift pit generally consists of RCC walls, a base slab and a PCC layer below the base. Proper dimensions, reinforcement, levels and waterproofing arrangements are important for safe and effective construction.

This project focuses on the preparation and interpretation of an RCC Lift Pit drawing using AutoCAD. The drawing includes the Lift Pit Plan and Section X-X′, showing important details such as RCC wall thickness, base slab, PCC, ground level, finished floor level, dimensions and reinforcement details. AutoCAD drawing and modification commands are used to prepare an accurate and well-organized 2D drawing.

The project provides practical knowledge of Civil Engineering drafting, RCC detailing, dimensioning, hatching, annotation and plotting

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2. INTRODUCTION

An RCC Lift Pit is an important structural component provided below the lowest floor level of a building. It provides the required space for lift equipment such as buffers, guide rails and other associated components.

The lift pit is generally constructed using reinforced cement concrete. The main components include RCC walls, RCC base slab and PCC below the base slab. Reinforcement is provided according to the approved structural design.

In this project, the RCC Lift Pit Plan and Section X-X′ are prepared and studied using AutoCAD. The drawing shows dimensions, levels, RCC wall thickness, PCC, reinforcement and other construction details.

AutoCAD helps Civil Engineers prepare accurate, editable and easily printable construction drawings.

3. AIM

To prepare and understand the RCC Lift Pit Plan and Section X-X′ using AutoCAD with accurate dimensions, RCC detailing, reinforcement indications and proper annotations.

4. OBJECTIVES

To understand the purpose of an RCC Lift Pit.

To study the Lift Pit Plan and Section X-X′.

To understand RCC walls, base slab and PCC.

To understand the dimensions and levels shown in the drawing.

To learn AutoCAD drawing commands.

To learn AutoCAD modification commands.

To prepare the lift pit plan accurately.

To prepare the section of the lift pit.

To represent reinforcement details.

To apply dimensions and annotations.

To use layers, line types and hatching properly.

To develop practical Civil Engineering drafting skills.

To understand the application of the drawing at construction sites.

To prepare a professional drawing suitable for printing and submission.

5. DRAWING INTERPRETATION

The supplied RCC Lift Pit drawing contains two major views:

A. Lift Pit Plan

The plan represents the horizontal arrangement of the lift pit.

Important dimensions shown are:

Overall dimension = 4200 × 4200 mm

Reference/internal dimension = 4000 × 4000 mm

RCC wall thickness = 230 mm

Offset = 600 mm

Section line = X-X′

Lift identification = F5 LIFT PIT

B. Section X-X′

The section represents the vertical arrangement of the lift pit.

Important details shown are:

RCC wall thickness = 230 mm

RCC base slab thickness = 450 mm

PCC thickness = 150 mm

Offset = 600 mm

Edge dimension = 75 mm

Vertical dimensions shown = 1000 mm and 2000 mm

Ground Level = G.L.

Finished Floor Level = F.F.L.

Reinforcement shown

8Ø @ 200 c/c

10Ø @ 200 c/c

10Ø @ 125 c/c

6. AUTOCAD DRAWING TOOLS

Command

Application

LINE (L)

Drawing walls, slabs and reference lines

POLYLINE (PL)

Creating continuous boundaries

RECTANGLE (REC)

Creating 4200 × 4200 mm plan

OFFSET (O)

Creating 230 mm wall thickness and offsets

CIRCLE (C)

Representing reinforcement dots/symbols

ARC (A)

Drawing curved details if required

HATCH (H)

Showing RCC and PCC in section

XLINE (XL)

Construction and projection lines

MTEXT (MT)

Writing drawing notes

DIMENSION

Adding dimensions

MLEADER

Adding reinforcement and component call-outs

7. AUTOCAD MODIFY TOOLS

Command

Application

ERASE (E)

Delete unwanted objects

MOVE (M)

Move objects

COPY (CO)

Copy similar details

MIRROR (MI)

Create symmetrical details

ROTATE (RO)

Rotate objects

TRIM (TR)

Remove unwanted portions

EXTEND (EX)

Extend lines

STRETCH (S)

Adjust dimensions/geometry

SCALE (SC)

Change object size

FILLET (F)

Modify corners

PEDIT (PE)

Edit polylines

JOIN (J)

Join lines

MATCHPROP (MA)

Copy object properties

EXPLODE (X)

Separate combined objects

8. DRAWING PROCEDURE

Open AutoCAD.

Set Lift Pit Plan

units to millimetres.

Create required layers.

Draw the 4200 × 4200 mm outer reference square.

Draw the 4000 × 4000 mm reference/internal square.

Use OFFSET to create the 230 mm RCC wall.

Mark the 600 mm offset as shown in the drawing.

Use TRIM and EXTEND to clean the geometry.

Mark the X-X′ section line.

Add F5 LIFT PIT text.

Add dimensions.

Add reinforcement indications.

Apply hatch where required.

Check all dimensions before finalizing.

9. SECTION X-X′ PROCEDURE

Project the plan geometry vertically.

Draw the G.L. and F.F.L. reference lines.

Mark the required vertical dimensions.

Draw the 230 mm RCC wall.

Draw the 450 mm RCC base slab.

Draw the 150 mm PCC layer below the base.

Show the 600 mm offset.

Add the 75 mm edge dimension where shown.

Apply RCC and PCC hatch.

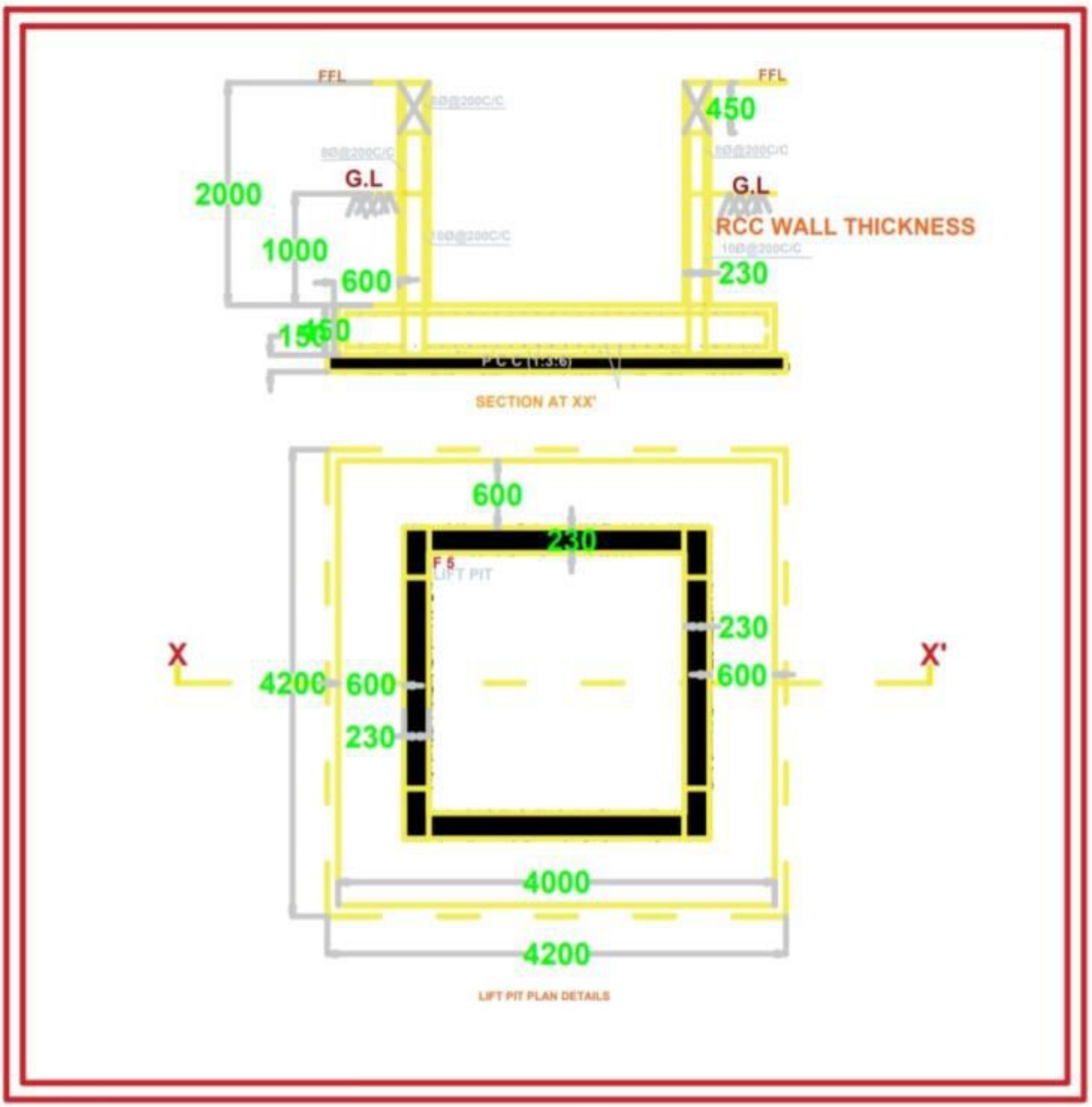
Add reinforcement notes.

Add leaders for RCC wall thickness.

Add GL and FFL labels.

Add section title.

Check the section against the plan.


FFL
FFL
450
G.L
G.L
2000
1000
600
RCC WALL THICKNESS
230
150
P.C.C (1:3:6)
SECTION AT XX'
600
230
F 5
LIFT PIT
230
X
X'
4200
600
600
230
4000
4200
LIFT PIT PLAN DETAILS

10. DIMENSIONING AND ANNOTATION

Proper dimensioning is necessary to make the drawing easy to understand.

Important dimensions include:

4200 mm

4000 mm

230 mm

600 mm

450 mm

150 mm

75 mm

1000 mm

2000 mm

Important annotations include:

F5 LIFT PIT

RCC WALL THICKNESS

G.L.

F.F.L.

P.C.C. (1:3:6)

8Ø @ 200 c/c

10Ø @ 200 c/c

10Ø @ 125 c/c

11. APPLICATIONS OF RCC LIFT PIT DRAWING

Construction Planning – Used for planning lift pit construction.

Site Execution – Helps site engineers understand dimensions and levels.

Excavation Work – Helps establish the required pit layout.

RCC Work – Provides wall and base slab information.

Reinforcement Work – Shows reinforcement arrangement and spacing.

PCC Work – Indicates the PCC layer below the RCC base.

Level Checking – Helps establish GL and FFL.

Quantity Estimation – Useful for estimating RCC, PCC, steel and formwork.

Coordination – Helps coordinate structural, architectural and lift-vendor requirements.

Documentation – CAD drawings can be revised, printed and archived.

As-Built Drawings – Can be updated after construction for project records.

Academic Training – Useful for learning Civil Engineering AutoCAD drafting.

12. ADVANTAGES

Provides accurate dimensions.

Easy to modify.

Reduces manual drafting errors.

Easy to print and share.

Helps in quantity estimation.

Improves construction coordination.

Provides clear representation of RCC details.

Can be stored for future reference.

Helps prepare professional engineering drawings.

13. SITE CONSIDERATIONS

Before construction, the following should be checked:

Approved structural drawing.

Lift manufacturer's requirements.

Lift pit depth and dimensions.

RCC wall thickness.

Base slab thickness.

Reinforcement details.

Concrete grade.

Reinforcement cover.

Waterproofing arrangement.

Drainage/sump arrangement.

Groundwater conditions.

Excavation safety.

Coordination with foundation and lift shaft.

14. CONCLUSION

The RCC Lift Pit AutoCAD project provides practical knowledge of preparing and interpreting a Civil Engineering construction drawing. The project covers the lift pit plan, Section X-X′, dimensions, RCC wall, base slab, PCC, reinforcement and level details.

By using AutoCAD drawing and modification commands, an accurate and editable 2D drawing can be prepared. The project also demonstrates the importance of layers, dimensions, hatching, text, leaders and plotting in professional CAD drafting.

The knowledge gained from this project can be applied to construction drawings, site execution, quantity estimation and coordination between different engineering disciplines.

15. REFERENCES

Supplied RCC Lift Pit Plan and Section X-X′ drawing.

Autodesk AutoCAD user documentation.

IS 456 – Plain and Reinforced Concrete – Code of Practice, where applicable.

Approved structural and architectural project drawings.

Lift manufacturer's technical drawings and specifications.

Civil Engineering AutoCAD drafting laboratory notes.

Note: The dimensions and reinforcement values in this report are taken from your supplied reference drawing. For actual construction, the latest approved structural and lift-vendor drawings must be followed.